

EQ Check Table

3rd October 2025 at 6:53pm

EO	Latex	KT	MP
AKolbe2018_EQ0001_0007		$\begin{aligned} c(\phi\rangle + \psi\rangle) &= c \phi\rangle + c \psi\rangle \\ (c + d) \phi\rangle &= c \phi\rangle + d \phi\rangle \\ c(d \phi\rangle) &= cd \phi\rangle \end{aligned}$	$\begin{aligned} c(\phi\rangle + \psi\rangle) &= c \phi\rangle + c \psi\rangle \\ (c + d) \phi\rangle &= c \phi\rangle + d \phi\rangle \\ c(d \phi\rangle) &= cd \phi\rangle \end{aligned}$
AKolbe2018_EQ0002_0008		$\vec{J} = \vec{r} \times \vec{p}$	$\vec{J} = \vec{r} \times \vec{p}$
AKolbe2018_EQ0003_0008		$\hat{J} = \frac{i}{\hbar} \hat{r} \times \nabla$	$\hat{J} = \frac{i}{\hbar} \hat{r} \times \nabla$
AKolbe2018_EQ0004_0008		$\hat{J}_x = \frac{i}{\hbar} \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \quad ; \quad \hat{J}_y = \frac{i}{\hbar} \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \quad ; \quad \hat{J}_z = \frac{i}{\hbar} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$	$\hat{J}_x = \frac{i}{\hbar} \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \quad ; \quad \hat{J}_y = \frac{i}{\hbar} \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \quad ; \quad \hat{J}_z = \frac{i}{\hbar} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$
AKolbe2018_EQ0005_0008		$[\hat{J}_x, \hat{J}_y] = i\hbar \hat{J}_z \quad ; \quad [\hat{J}_y, \hat{J}_z] = i\hbar \hat{J}_x \quad ; \quad [\hat{J}_z, \hat{J}_x] = i\hbar \hat{J}_y$	$[\hat{J}_x, \hat{J}_y] = i\hbar \hat{J}_z \quad ; \quad [\hat{J}_y, \hat{J}_z] = i\hbar \hat{J}_x \quad ; \quad [\hat{J}_z, \hat{J}_x] = i\hbar \hat{J}_y$
AKolbe2018_EQ0006_0009		$\epsilon_{ijk} = \begin{cases} +1, & \text{wenn } ijk \text{ eine gerade Permutation von } 123 \text{ ist} \\ -1, & \text{wen } ijk \text{ eine ungerade Permutation von } 123 \text{ ist} \\ 0, & \text{wenn mindestens zwei Indizes gleich sind} \end{cases}$	$\epsilon_{ijk} = \begin{cases} +1, & \text{wenn } ijk \text{ eine gerade Permutation von } 123 \text{ ist} \\ -1, & \text{wen } ijk \text{ eine ungerade Permutation von } 123 \text{ ist} \\ 0, & \text{wenn mindestens zwei Indizes gleich sind} \end{cases}$

AKolbe2018_EQ0007_0009		$\left[\hat{J}_i, \hat{J}_j\right] = \mathrm{i} \hbar \sum_k \hat{J}_k \epsilon_{ijk}$	$\left[\hat{J}_i, \hat{J}_j\right] = \mathrm{i} \hbar \sum_k \hat{J}_k \epsilon_{ijk}$
AKolbe2018_EQ0008_0009		$\left[\hat{J}_i, \hat{J}^2\right] = 0 \quad i = x, y, z$	$\left[\hat{J}_i, \hat{J}^2\right] = 0 \quad i = x, y, z$
AKolbe2018_EQ0009_0009		$\begin{aligned} \hat{\mathbf{J}}^2 j, m\rangle &= \hbar^2 j(j+1) j, m\rangle \\ \hat{J}_z j, m\rangle &= \hbar m j, m\rangle \end{aligned}$	$\begin{aligned} \hat{\mathbf{J}}^2 j, m\rangle &= \hbar^2 j(j+1) j, m\rangle \\ \hat{J}_z j, m\rangle &= \hbar m j, m\rangle \end{aligned}$
AKolbe2018_EQ0010_0009		$\begin{aligned} \hat{J}^+ &= \left(\hat{J}_x + \mathrm{i} \hat{J}_y\right) \\ \hat{J}^- &= \left(\hat{J}_x - \mathrm{i} \hat{J}_y\right) \end{aligned}$	$\begin{aligned} \hat{J}^+ &= \left(\hat{J}_x + \mathrm{i} \hat{J}_y\right) \\ \hat{J}^- &= \left(\hat{J}_x - \mathrm{i} \hat{J}_y\right) \end{aligned}$
AKolbe2018_EQ0011_0009		$\left[\hat{J}_z, \hat{J}^+\right] = \hbar \hat{J}^+ \quad , \quad \left[\hat{J}_z, \hat{J}^-\right] = -\hbar \hat{J}^- \quad , \quad \left[\hat{J}^+, \hat{J}^-\right] = 2 \hbar \hat{J}_z$	$\left[\hat{J}_z, \hat{J}^+\right] = \hbar \hat{J}^+ \quad , \quad \left[\hat{J}_z, \hat{J}^-\right] = -\hbar \hat{J}^- \quad , \quad \left[\hat{J}^+, \hat{J}^-\right] = 2 \hbar \hat{J}_z$
AKolbe2018_EQ0012_0010		$ s\rangle = \alpha \uparrow\rangle + \beta \downarrow\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$	$ s\rangle = \alpha \uparrow\rangle + \beta \downarrow\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$
AKolbe2018_EQ0013_0010		$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad ; \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} \quad ; \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad ; \quad \hat{\sigma}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad ; \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} \quad ; \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad ; \quad \hat{\sigma}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

AKolbe2018_EQ0014_0010		$\vec{S} = \frac{\hbar}{2} \vec{\sigma}$ $\hat{S}_i = \frac{\hbar}{2} \hat{\sigma}_i \quad i = x, y, z$	$\vec{S} = \frac{\hbar}{2} \vec{\sigma}$ $\hat{S}_i = \frac{\hbar}{2} \hat{\sigma}_i \quad i = x, y, z$
AKolbe2018_EQ0015_0010		$\mathbf{1}_2 = \hat{\sigma}_x^2 + \hat{\sigma}_y^2 + \hat{\sigma}_z^2$ $2i\hat{\sigma}_z = [\hat{\sigma}_x, \hat{\sigma}_y] \text{ und zyklisch}$ $\{\hat{\sigma}_x, \hat{\sigma}_y\} = \{\hat{\sigma}_y, \hat{\sigma}_z\} = \{\hat{\sigma}_z, \hat{\sigma}_x\} = 0$	$\mathbf{1}_2 = \hat{\sigma}_x^2 + \hat{\sigma}_y^2 + \hat{\sigma}_z^2$ $2i\hat{\sigma}_z = [\hat{\sigma}_x, \hat{\sigma}_y] \text{ und zyklisch}$ $\{\hat{\sigma}_x, \hat{\sigma}_y\} = \{\hat{\sigma}_y, \hat{\sigma}_z\} = \{\hat{\sigma}_z, \hat{\sigma}_x\} = 0$
AKolbe2018_EQ0016_0011		$\hat{S}_x = \frac{1}{2} (\hat{S}^+ + \hat{S}^-)$ $\hat{S}_y = \frac{1}{2i} (\hat{S}^+ - \hat{S}^-)$	$\hat{S}_x = \frac{1}{2} (\hat{S}^+ + \hat{S}^-)$ $\hat{S}_y = \frac{1}{2i} (\hat{S}^+ - \hat{S}^-)$
AKolbe2018_EQ0017_0011		$\hat{S}^+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad ; \quad \hat{S}^- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$	$\hat{S}^+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad ; \quad \hat{S}^- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$
AKolbe2018_EQ0018_0011		$\hat{S}^+ \downarrow\rangle = \hbar \uparrow\rangle \quad \hat{S}^+ \uparrow\rangle = 0$ $\hat{S}^- \uparrow\rangle = \hbar \downarrow\rangle \quad \hat{S}^- \downarrow\rangle = 0$ $\hat{S}_z \uparrow\rangle = \frac{\hbar}{2} \uparrow\rangle \quad \hat{S}_z \downarrow\rangle = -\frac{\hbar}{2} \downarrow\rangle$	$\hat{S}^+ \downarrow\rangle = \hbar \uparrow\rangle \quad \hat{S}^+ \uparrow\rangle = 0$ $\hat{S}^- \uparrow\rangle = \hbar \downarrow\rangle \quad \hat{S}^- \downarrow\rangle = 0$ $\hat{S}_z \uparrow\rangle = \frac{\hbar}{2} \uparrow\rangle \quad \hat{S}_z \downarrow\rangle = -\frac{\hbar}{2} \downarrow\rangle$

AKolbe2018_EQ0019_0011		$\begin{aligned} -1 < \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} < 0 & \text{ antikorreliert.} \\ 0 < \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} < 1 & \text{ korreliert.} \\ 0 = \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} & \text{ unkorreliert.} \\ 1 = \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} & \text{ maximal korreliert.} \end{aligned}$	$\begin{aligned} -1 < \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} < 0 & \text{ antikorreliert.} \\ 0 < \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} < 1 & \text{ korreliert.} \\ 0 = \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} & \text{ unkorreliert.} \\ 1 = \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} & \text{ maximal korreliert.} \end{aligned}$
AKolbe2018_EQ0020_0011		$\langle \vec{S}_i \vec{S}_j \rangle_T = \frac{1}{Z} \sum_n e^{\beta E_n} \sum_{l=1}^{g_n} \langle \Psi_{E_n,l} \vec{S}_i \vec{S}_j \Psi_{E_n,l} \rangle$	$\langle \vec{S}_i \vec{S}_j \rangle_T = \frac{1}{Z} \sum_n e^{\beta E_n} \sum_{l=1}^{g_n} \langle \Psi_{E_n,l} \vec{S}_i \vec{S}_j \Psi_{E_n,l} \rangle$
AKolbe2018_EQ0021_0011		$\langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} = \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \hat{S}_i \hat{S}_j \psi^{0,l} \rangle$	$\langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} = \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \hat{S}_i \hat{S}_j \psi^{0,l} \rangle$
AKolbe2018_EQ0022_0012		$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_{N-1} \otimes \mathcal{H}_N$	$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_{N-1} \otimes \mathcal{H}_N$
AKolbe2018_EQ0023_0012		$ \psi\rangle = \psi\rangle_1 \otimes \psi\rangle_2 \otimes \dots \otimes \psi\rangle_{N-1} \otimes \psi\rangle_N$	$ \psi\rangle = \psi\rangle_1 \otimes \psi\rangle_2 \otimes \dots \otimes \psi\rangle_{N-1} \otimes \psi\rangle_N$
AKolbe2018_EQ0024_0012		$\hat{A}_i : \mathcal{H}_i \rightarrow \mathcal{H}_i$	$\hat{A}_i : \mathcal{H}_i \rightarrow \mathcal{H}_i$
AKolbe2018_EQ0025_0012		$\overline{A}_1 + \dots + \overline{A}_N = \hat{A}_1 \otimes \hat{I} \otimes \dots \otimes \hat{I} + \hat{I} \otimes \hat{A}_2 \otimes \dots \otimes \hat{I} + \dots + \hat{I} \otimes \dots \otimes \hat{A}_N$	$\overline{A}_1 + \dots + \overline{A}_N = \hat{A}_1 \otimes \hat{I} \otimes \dots \otimes \hat{I} + \hat{I} \otimes \hat{A}_2 \otimes \dots \otimes \hat{I} + \dots + \hat{I} \otimes \dots \otimes \hat{A}_N$
AKolbe2018_EQ0026_0013		$\mathcal{H} = \mathcal{H}_1^{(1)} \otimes \mathcal{H}_1^{(2)} \otimes \dots \otimes \mathcal{H}_1^{(N)}$	$\mathcal{H} = \mathcal{H}_1^{(1)} \otimes \mathcal{H}_1^{(2)} \otimes \dots \otimes \mathcal{H}_1^{(N)}$
AKolbe2018_EQ0027_0013		$\mathcal{V}(x_1, x_2, \dots, x_N) = \sum_{\{\alpha_i\}} a_{\alpha_1 \alpha_2 \dots \alpha_N} \nu_{\alpha_1}(x_1) \nu_{\alpha_2}(x_2) \dots \nu_{\alpha_N}(x_N), \quad a_{\alpha_i} \in \mathbb{C}$	$\mathcal{V}(x_1, x_2, \dots, x_N) = \sum_{\{\alpha_i\}} a_{\alpha_1 \alpha_2 \dots \alpha_N} \nu_{\alpha_1}(x_1) \nu_{\alpha_2}(x_2) \dots \nu_{\alpha_N}(x_N), \quad a_{\alpha_i} \in \mathbb{C}$

AKolbe2018_EQ0028_0013		$\begin{aligned}\Psi(x_1, x_2, \dots, x_N) &= \text{sign}(\pi) \Psi(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}) && \text{für Fermionen} \\ \Phi(x_1, x_2, \dots, x_N) &= \Phi(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}) && \text{für Bosonen}\end{aligned}$	$\begin{aligned}\Psi(x_1, x_2, \dots, x_N) &= \text{sign}(\pi) \Psi(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}) && \text{für Fermionen} \\ \Phi(x_1, x_2, \dots, x_N) &= \Phi(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}) && \text{für Bosonen}\end{aligned}$
AKolbe2018_EQ0029_0013		$\Psi(x_1, x_2, \dots, x_N) = \sum_{\{\alpha_i\}} a_{\alpha_1 \alpha_2 \dots \alpha_N} \psi_{\alpha_1 \alpha_2 \dots \alpha_N}(x_1, x_2, \dots, x_N), \quad a_{\alpha_1 \alpha_2 \dots \alpha_N} \in \mathbb{C}$	$\Psi(x_1, x_2, \dots, x_N) = \sum_{\{\alpha_i\}} a_{\alpha_1 \alpha_2 \dots \alpha_N} \psi_{\alpha_1 \alpha_2 \dots \alpha_N}(x_1, x_2, \dots, x_N), \quad a_{\alpha_1 \alpha_2 \dots \alpha_N} \in \mathbb{C}$
AKolbe2018_EQ0030_0013		$\psi_{\alpha_1 \alpha_2 \dots \alpha_N}(x_1, x_2, \dots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \psi_{\alpha_1}(x_1) & \dots & \psi_{\alpha_N}(x_1) \\ \vdots & \ddots & \vdots \\ \psi_{\alpha_1}(x_N) & \dots & \psi_{\alpha_N}(x_N) \end{pmatrix}$	$\psi_{\alpha_1 \alpha_2 \dots \alpha_N}(x_1, x_2, \dots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \psi_{\alpha_1}(x_1) & \dots & \psi_{\alpha_N}(x_1) \\ \vdots & \ddots & \vdots \\ \psi_{\alpha_1}(x_N) & \dots & \psi_{\alpha_N}(x_N) \end{pmatrix}$
AKolbe2018_EQ0031_0014		$\langle n_{1\sigma}, n_{2\sigma}, \dots \mid n'_{1\sigma}, n'_{2\sigma}, \dots \rangle = \prod_{i,\sigma} \delta_{n_{i\sigma}, n'_{i\sigma}}$	$\langle n_{1\sigma}, n_{2\sigma}, \dots \mid n'_{1\sigma}, n'_{2\sigma}, \dots \rangle = \prod_{i,\sigma} \delta_{n_{i\sigma}, n'_{i\sigma}}$
AKolbe2018_EQ0032_0014		$\lvert \Psi \rangle = \sum_{n_{1\sigma}, n_{2\sigma}, \dots} a_{n_{1\sigma}, n_{2\sigma}, \dots} \lvert n_{1\sigma}, n_{2\sigma}, \dots \rangle, \quad \text{wobei } 2N = \sum_{i,\sigma} n_{i,\sigma}$	$\lvert \Psi \rangle = \sum_{n_{1\sigma}, n_{2\sigma}, \dots} a_{n_{1\sigma}, n_{2\sigma}, \dots} \lvert n_{1\sigma}, n_{2\sigma}, \dots \rangle, \quad \text{wobei } 2N = \sum_{i,\sigma} n_{i,\sigma}$
AKolbe2018_EQ0033_0014		$F^{(2N)} = \mathcal{H}^{(+)} = \{ \lvert \Psi \rangle \in \mathcal{H} \mid \hat{\pi} \lvert \Psi \rangle = (-1)^\pi \lvert \Psi \rangle, \forall \pi \in S_n \}$	$F^{(2N)} = \mathcal{H}^{(+)} = \{ \lvert \Psi \rangle \in \mathcal{H} \mid \hat{\pi} \lvert \Psi \rangle = (-1)^\pi \lvert \Psi \rangle, \forall \pi \in S_n \}$
AKolbe2018_EQ0034_0014		$c_{i\sigma}^\dagger \lvert n_{1\sigma}, \dots, n_{i\sigma}, \dots \rangle = \left(1 - n_{i\sigma} (-1)^{\sum_{\sigma, j < i} n_{j\sigma}} \right) \lvert n_{1\sigma}, \dots, n_{i\sigma} + 1, \dots \rangle$	$c_{i\sigma}^\dagger \lvert n_{1\sigma}, \dots, n_{i\sigma}, \dots \rangle = \left(1 - n_{i\sigma} (-1)^{\sum_{\sigma, j < i} n_{j\sigma}} \right) \lvert n_{1\sigma}, \dots, n_{i\sigma} + 1, \dots \rangle$
AKolbe2018_EQ0035_0014		$\lvert n_{1\uparrow} n_{1\downarrow} \dots \rangle = \prod_{i,\sigma} \frac{1}{\sqrt{n_{i\sigma}!}} \left(c_{i,\sigma}^\dagger \right)^{n_{i\sigma}} \lvert 0 \rangle$	$\lvert n_{1\uparrow} n_{1\downarrow} \dots \rangle = \prod_{i,\sigma} \frac{1}{\sqrt{n_{i\sigma}!}} (c_{i,\sigma}^\dagger)^{n_{i\sigma}} \lvert 0 \rangle$

AKolbe2018_EQ0036_0015		$c_{i\sigma} \left n_{1\sigma}, \dots, n_{i\sigma}, \dots \right\rangle = n_{i\sigma} (-1)^{\sum_{\sigma, j < i} 2n_{j\sigma}} \left n_{1\sigma}, \dots, n_{i\sigma} - 1, \dots \right\rangle$	$c_{i\sigma} \left n_{1\sigma}, \dots, n_{i\sigma}, \dots \right\rangle = n_{i\sigma} (-1)^{\sum_{\sigma, j < i} 2n_{j\sigma}} \left n_{1\sigma}, \dots, n_{i\sigma} - 1, \dots \right\rangle$
AKolbe2018_EQ0037_0015		$\begin{aligned} [c_{i\sigma}, c_{j\sigma'}^\dagger] &= \delta_{ij} \delta_{\sigma\sigma'} \\ [c_{i\sigma}, c_{j\sigma'}] &= 0 \\ [c_{i\sigma}^\dagger, c_{j\sigma'}^\dagger] &= 0 \end{aligned}$	$\begin{aligned} [c_{i\sigma}, c_{j\sigma'}^\dagger] &= \delta_{ij} \delta_{\sigma\sigma'} \\ [c_{i\sigma}, c_{j\sigma'}] &= 0 \\ [c_{i\sigma}^\dagger, c_{j\sigma'}^\dagger] &= 0 \end{aligned}$
AKolbe2018_EQ0038_0015		$\hat{n}_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$	$\hat{n}_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$
AKolbe2018_EQ0039_0015		$\hat{n}_{i\sigma} \left n_{1\sigma}, \dots, n_{i\sigma}, \dots \right\rangle = n_{i\sigma} \left n_{1\sigma}, \dots, n_{i\sigma}, \dots \right\rangle$	$\hat{n}_{i\sigma} \left n_{1\sigma}, \dots, n_{i\sigma}, \dots \right\rangle = n_{i\sigma} \left n_{1\sigma}, \dots, n_{i\sigma}, \dots \right\rangle$
AKolbe2018_EQ0040_0017		$\mathbf{H}_{\text{Heis}} = -J \sum_{\langle i, j \rangle} \vec{S}_i \cdot \vec{S}_j$	$\mathbf{H}_{\text{Heis}} = -J \sum_{\langle i, j \rangle} \vec{S}_i \cdot \vec{S}_j$
AKolbe2018_EQ0041_0017		$\vec{S}_i \cdot \vec{S}_j = S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z$	$\vec{S}_i \cdot \vec{S}_j = S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z$
AKolbe2018_EQ0042_0017		$\mathbf{H}_{\text{Heis}} = -J \sum_{\langle i, j \rangle} \frac{\hat{S}_i^+ \hat{S}_j^-}{2} + \frac{\hat{S}_i^- \hat{S}_j^+}{2} + \hat{S}_i^z \hat{S}_j^z$	$\mathbf{H}_{\text{Heis}} = -J \sum_{\langle i, j \rangle} \frac{\hat{S}_i^+ \hat{S}_j^-}{2} + \frac{\hat{S}_i^- \hat{S}_j^+}{2} + \hat{S}_i^z \hat{S}_j^z$
AKolbe2018_EQ0043_0019		$\mathbf{H}_{\text{Hub}} = \mathbf{H}_U + \mathbf{H}_\epsilon + \mathbf{H}_t$	$\mathbf{H}_{\text{Hub}} = \mathbf{H}_U + \mathbf{H}_\epsilon + \mathbf{H}_t$
AKolbe2018_EQ0044_0019		$\mathbf{H}_{\text{Hub}} = U \sum_{i=1}^N \hat{c}_{i,\uparrow}^\dagger \hat{c}_{i,\uparrow} \hat{c}_{i,\downarrow}^\dagger \hat{c}_{i,\downarrow} + \sum_{\sigma} \left(\sum_{i=1}^N \epsilon_i \hat{c}_{i,\sigma}^\dagger \hat{c}_{i,\sigma} - t \sum_{\langle i, j \rangle} \left(\hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} + \hat{c}_{j,\sigma}^\dagger \hat{c}_{i,\sigma} \right) \right)$	$\mathbf{H}_{\text{Hub}} = U \sum_{i=1}^N \hat{c}_{i,\uparrow}^\dagger \hat{c}_{i,\uparrow} \hat{c}_{i,\downarrow}^\dagger \hat{c}_{i,\downarrow} + \sum_{\sigma} \left(\sum_{i=1}^N \epsilon_i \hat{c}_{i,\sigma}^\dagger \hat{c}_{i,\sigma} - t \sum_{\langle i, j \rangle} \left(\hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} + \hat{c}_{j,\sigma}^\dagger \hat{c}_{i,\sigma} \right) \right)$

AKolbe2018_EQ0045_0021		$\hat{S}_i^+ \hat{S}_j^- Zustand\rangle \equiv \text{SpinPlus}(i, \text{SpinMinus}(j, Zustand))$	$\hat{S}_i^+ \hat{S}_j^- Zustand\rangle \equiv \text{SpinPlus}(i, \text{SpinMinus}(j, Zustand))$
AKolbe2018_EQ0046_0021		$\begin{pmatrix} 0 & J & 0 \\ 0 & 0 & J \\ J & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & J & 0 \\ 0 & 0 & J \\ J & 0 & 0 \end{pmatrix}$
AKolbe2018_EQ0047_0021		$\langle \hat{S}_1 \hat{S}_j \rangle_{\psi_g} = \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{\hat{S}_1^+ \hat{S}_j^-}{2} + \frac{\hat{S}_1^- \hat{S}_j^+}{2} + \hat{S}_1^z \hat{S}_j^z \psi^{0,l} \rangle$	$\langle \hat{S}_1 \hat{S}_j \rangle_{\psi_g} = \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{\hat{S}_1^+ \hat{S}_j^-}{2} + \frac{\hat{S}_1^- \hat{S}_j^+}{2} + \hat{S}_1^z \hat{S}_j^z \psi^{0,l} \rangle$
AKolbe2018_EQ0048_0022		$ \psi_g\rangle = \sum_{k=1}^{2^N} \alpha_k k\rangle \quad \text{Heisenberg-Modell}$	$ \psi_g\rangle = \sum_{k=1}^{2^N} \alpha_k k\rangle \quad \text{Heisenberg-Modell}$
AKolbe2018_EQ0049_0023		$\hat{n}_{\uparrow} Zustand\rangle = \hat{c}_{i\uparrow}^\dagger \hat{c}_{j\uparrow} Zustand\rangle \Leftrightarrow \text{CupDagger}(i, \text{Cup}(j, Zustand))$	$\hat{n}_{\uparrow} Zustand\rangle = \hat{c}_{i\uparrow}^\dagger \hat{c}_{j\uparrow} Zustand\rangle \Leftrightarrow \text{CupDagger}(i, \text{Cup}(j, Zustand))$
AKolbe2018_EQ0050_0023		$\begin{pmatrix} 0 & t & 0 \\ 0 & 0 & t \\ 0 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & t & 0 \\ 0 & 0 & t \\ 0 & 0 & 0 \end{pmatrix}$
AKolbe2018_EQ0051_0023		$\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow}^\dagger 0\rangle_j = \uparrow\downarrow\rangle_j \quad \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow}^\dagger 0\rangle_j = -\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow}^\dagger 0\rangle_j = - \uparrow\downarrow\rangle_j$	$\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow}^\dagger 0\rangle_j = \uparrow\downarrow\rangle_j \quad \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow}^\dagger 0\rangle_j = -\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow}^\dagger 0\rangle_j = - \uparrow\downarrow\rangle_j$
AKolbe2018_EQ0052_0023		$ 0,0\rangle = [0,0] \quad \downarrow,\downarrow\rangle = [1,1] \quad \uparrow,\uparrow\rangle = [2,2] \quad \uparrow\downarrow,\uparrow\downarrow\rangle = [3,3]$	$ 0,0\rangle = [0,0] \quad \downarrow,\downarrow\rangle = [1,1] \quad \uparrow,\uparrow\rangle = [2,2] \quad \uparrow\downarrow,\uparrow\downarrow\rangle = [3,3]$

AKolbe2018_EQ0053_0023		$ \begin{aligned} 0, \downarrow\rangle &= [0, 1] \rightarrow \downarrow, 0\rangle = [1, 0] \\ \downarrow, 0\rangle &= [1, 0] \rightarrow 0, \downarrow\rangle = [0, 1] \\ 0, \uparrow\rangle &= [0, 2] \rightarrow \uparrow, 0\rangle = [2, 0] \\ \uparrow, 0\rangle &= [2, 0] \rightarrow 0, \uparrow\rangle = [0, 2] \end{aligned} $	$ \begin{aligned} 0, \downarrow\rangle &= [0, 1] \rightarrow \downarrow, 0\rangle = [1, 0] \\ \downarrow, 0\rangle &= [1, 0] \rightarrow 0, \downarrow\rangle = [0, 1] \\ 0, \uparrow\rangle &= [0, 2] \rightarrow \uparrow, 0\rangle = [2, 0] \\ \uparrow, 0\rangle &= [2, 0] \rightarrow 0, \uparrow\rangle = [0, 2] \end{aligned} $
AKolbe2018_EQ0054_0024		$ \begin{aligned} \uparrow, \uparrow\downarrow\rangle &= [1, 3] \rightarrow \uparrow\downarrow, \uparrow\rangle = [3, 1] \\ \downarrow, \downarrow\uparrow\rangle &= [1, 3] \rightarrow \uparrow\downarrow, \downarrow\rangle = [3, 2] \\ \uparrow\downarrow, \uparrow\rangle &= [3, 1] \rightarrow \uparrow, \uparrow\downarrow\rangle = [1, 3] \\ \uparrow\downarrow, \downarrow\rangle &= [3, 2] \rightarrow \downarrow, \uparrow\downarrow\rangle = [2, 3] \end{aligned} $	$ \begin{aligned} \uparrow, \uparrow\downarrow\rangle &= [1, 3] \rightarrow \uparrow\downarrow, \uparrow\rangle = [3, 1] \\ \downarrow, \downarrow\uparrow\rangle &= [1, 3] \rightarrow \uparrow\downarrow, \downarrow\rangle = [3, 2] \\ \uparrow\downarrow, \uparrow\rangle &= [3, 1] \rightarrow \uparrow, \uparrow\downarrow\rangle = [1, 3] \\ \uparrow\downarrow, \downarrow\rangle &= [3, 2] \rightarrow \downarrow, \uparrow\downarrow\rangle = [2, 3] \end{aligned} $
AKolbe2018_EQ0055_0024		$ \begin{aligned} \uparrow, \downarrow\rangle &= [1, 2] \rightarrow \uparrow\downarrow, 0\rangle + 0, \uparrow\downarrow\rangle \\ \downarrow, \uparrow\rangle &= [2, 1] \rightarrow - \uparrow\downarrow, 0\rangle - 0, \uparrow\downarrow\rangle \\ \uparrow\downarrow, 0\rangle &= [3, 0] \rightarrow \uparrow, \downarrow\rangle - \downarrow, \uparrow\rangle = [1, 2] - [2, 1] \\ \uparrow, 0\rangle &= [2, 1] - [1, 2] \\ 0, \uparrow\downarrow\rangle &= [0, 3] \rightarrow \uparrow, \downarrow\rangle - \downarrow, \uparrow\rangle \end{aligned} $	$ \begin{aligned} \uparrow, \downarrow\rangle &= [1, 2] \rightarrow \uparrow\downarrow, 0\rangle + 0, \uparrow\downarrow\rangle = [3, 0] + [0, 3] \\ \downarrow, \uparrow\rangle &= [2, 1] \rightarrow - \uparrow\downarrow, 0\rangle - 0, \uparrow\downarrow\rangle = -[3, 0] - [0, 3] \\ \uparrow\downarrow, 0\rangle &= [3, 0] \rightarrow \uparrow, \downarrow\rangle - \downarrow, \uparrow\rangle = [2, 1] - [1, 2] \\ 0, \uparrow\downarrow\rangle &= [0, 3] \rightarrow \uparrow, \downarrow\rangle - \downarrow, \uparrow\rangle = [1, 2] - [2, 1] \end{aligned} $
AKolbe2018_EQ0056_0024		$ \mathbf{H}_\epsilon + \mathbf{H}_U = \sum_{i=1}^N (\epsilon_i \cdot (\hat{n}_{i,\uparrow} + \hat{n}_{i,\downarrow}) + U_i \cdot \hat{n}_{i,\uparrow} \hat{n}_{i,\downarrow}) $	$ \mathbf{H}_\epsilon + \mathbf{H}_U = \sum_{i=1}^N (\epsilon_i \cdot (\hat{n}_{i,\uparrow} + \hat{n}_{i,\downarrow}) + U_i \cdot \hat{n}_{i,\uparrow} \hat{n}_{i,\downarrow}) $

AKolbe2018_EQ0057_0024		$\langle l \mathbf{H}_\epsilon + \mathbf{H}_U m \rangle = \begin{cases} \epsilon_{lm} + U_{lm} & \text{wenn } l = m \text{ ist.} \\ 0 & \text{wenn } l \neq m \text{ ist.} \end{cases}$	$\langle l \mathbf{H}_\epsilon + \mathbf{H}_U m \rangle = \begin{cases} \epsilon_{lm} + U_{lm} & \text{wenn } l = m \text{ ist.} \\ 0 & \text{wenn } l \neq m \text{ ist.} \end{cases}$
AKolbe2018_EQ0058_0025		$\hat{S}_i^+ = \hat{c}_{i,\uparrow}^\dagger \hat{c}_{i,\downarrow}; \quad \hat{S}_i^- = \hat{c}_{i,\downarrow}^\dagger \hat{c}_{i,\uparrow}; \quad \hat{S}_i^z = \frac{1}{2} \left(\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} - \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} \right)$	$\hat{S}_i^+ = \hat{c}_{i,\uparrow}^\dagger \hat{c}_{i,\downarrow} \ ; \ \hat{S}_i^- = \hat{c}_{i,\downarrow}^\dagger \hat{c}_{i,\uparrow} \ ; \ \hat{S}_i^z = \frac{1}{2} \left(\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} - \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} \right)$
AKolbe2018_EQ0059_0025		$\begin{aligned} C_{1j} &= \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{1}{2} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\downarrow} \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\uparrow} \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow} \right) \\ &\quad + \frac{1}{4} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\uparrow} - \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\downarrow} \right) \left(\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} - \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} \right) \psi^{0,l} \rangle \\ &= \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{1}{2} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\downarrow} \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\uparrow} \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow} \right) \\ &\quad + \frac{1}{4} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\uparrow} \cdot \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} - \hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\uparrow} \cdot \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} - \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\downarrow} \cdot \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\downarrow} \cdot \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} \right) \psi^{0,l} \rangle \end{aligned}$	$\begin{aligned} C_{1j} &= \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{1}{2} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\downarrow} \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\uparrow} \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow} \right) \\ &\quad + \frac{1}{4} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\uparrow} - \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\downarrow} \right) \left(\hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} - \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} \right) \psi^{0,l} \rangle \\ &= \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{1}{2} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\downarrow} \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\uparrow} \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow} \right) \\ &\quad + \frac{1}{4} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\uparrow} \cdot \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} - \hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\uparrow} \cdot \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} - \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\downarrow} \cdot \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\downarrow} \cdot \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\downarrow} \right) \psi^{0,l} \rangle \end{aligned}$
AKolbe2018_EQ0060_0025		$\begin{aligned} \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} &= \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{1}{2} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\downarrow} \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\uparrow} \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow} \right) \\ &\quad + \frac{1}{4} \left(\hat{n}_{1,\uparrow} \hat{n}_{j,\uparrow} - \hat{n}_{1,\uparrow} \hat{n}_{j,\downarrow} - \hat{n}_{1,\downarrow} \hat{n}_{j,\uparrow} + \hat{n}_{1,\downarrow} \hat{n}_{j,\downarrow} \right) \psi^{0,l} \rangle \end{aligned}$	$\begin{aligned} \langle \hat{S}_i \hat{S}_j \rangle_{\psi_g} &= \frac{1}{g_0} \sum_{l=0}^{g_0} \langle \psi^{0,l} \frac{1}{2} \left(\hat{c}_{1,\uparrow}^\dagger \hat{c}_{1,\downarrow} \hat{c}_{j,\downarrow}^\dagger \hat{c}_{j,\uparrow} + \hat{c}_{1,\downarrow}^\dagger \hat{c}_{1,\uparrow} \hat{c}_{j,\uparrow}^\dagger \hat{c}_{j,\downarrow} \right) \\ &\quad + \frac{1}{4} \left(\hat{n}_{1,\uparrow} \hat{n}_{j,\uparrow} - \hat{n}_{1,\uparrow} \hat{n}_{j,\downarrow} - \hat{n}_{1,\downarrow} \hat{n}_{j,\uparrow} + \hat{n}_{1,\downarrow} \hat{n}_{j,\downarrow} \right) \psi^{0,l} \rangle \end{aligned}$

AKolbe2018_EQ0061_0025		$ \psi_g\rangle = \sum_{k=1}^{4^N} \alpha_k k\rangle \quad \text{Hubbard-Modell}$	$ \psi_g\rangle = \sum_{k=1}^{4^N} \alpha_k k\rangle \quad \text{Hubbard-Modell}$
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